GTPA v1.2 → GTPA v2 — crosswalk

Status:

```
document  = crosswalk from GTPA v1.2 (DOI 10.5281/zenodo.17994303) to the GTPA v2
            text (gtpa-v2/GTPA-v2.md), and its errata annex / started 2026-09-26 /
            since 2026-09-26 every numbered v1.2 statement has a disposition
            here (§3, §4)
§1        = the 29 (+2) questions, all answered, in 34 verdict rows: Block A 5
            (copied from ufd/UFD-NUCLEO.md §3.7, pointing to the nucleus),
            Block B 8 (gtpa-v2/BLOCK-B-ANSWERS.md), Block C 21 = 18 questions
            + the Thm 9.1 half of EG-03.0 + the HOLDs EG-10.a2 and EG-09.5
            (gtpa-v2/BLOCK-C-THEOREMS.md, gtpa-v2/BLOCK-C-OBJECTS.md);
            plus one convention row (EG-09.1, EG-09.2)
tally §1  = 34 = 16 ILL_POSED_AS_WRITTEN / 15 FAILS_WITH_COUNTEREXAMPLE /
            3 HOLDS_UNDER_STATED_HYPOTHESES, plus the convention row
§2        = the v1.2 proof inventory, copied from the triage record
§3        = every numbered v1.2 statement: 71 rows (36 answered in a §1 row,
            12 short verdict in §4, 10 kept as a definition, 7 kept as a
            declared model, 5 dropped with the physics, 1 dropped as the
            definition of agency by the ledger)
tally §4  = 10 short verdicts (D-01..D-10) covering the 12 statements that no
            question asked: 7 ILL_POSED_AS_WRITTEN /
            2 FAILS_WITH_COUNTEREXAMPLE / 1 HOLDS_UNDER_STATED_HYPOTHESES
levels    = (1) Lean / (2) decisive calculation recomputed by machine /
            (3) independent AI read / (4) written only
checked   = §1 rows: every row read by an independent same-provider AI
            (level 3); on 2026-09-26 the decisive calculations behind 31 of
            the 34 verdicts were recomputed by machine and agreed (level 2,
            0 disagreements); EG-02.2, T6 and EG-09.5 rest on readings
            (level 3); column "Checked" in §1. §3-§4, written 2026-09-26:
            the decisive calculations of D-01, D-02 and D-06..D-10
            recomputed by machine once (level 2), D-03..D-05 have no machine
            computation (level 3 after the read below); one independent read of §3-§4 together with
            gtpa-v2/GTPA-v2.md (same provider, 2026-09-26) found two real
            errors, both fixed: D-04 stated its persistence condition as
            necessary (now the exact condition), and the EG-03.9 plain line
            stated a sufficient condition as necessary; one minor routing
            point (Def 2.2) applied; not re-read. The Block A recompute is
            recorded in the Checked column only; the UFD status block does
            not carry it
authority = a reading and routing table; claim authority stays in
            gtpa-lab/CLAIM-LEDGER.md; no row is a GTPA claim
verdicts  = HOLDS_UNDER_STATED_HYPOTHESES / FAILS_WITH_COUNTEREXAMPLE /
            ILL_POSED_AS_WRITTEN (+ the repair GTPA v2 adopts) / OPEN
physics = CLOSED     MATH<->GTPA = UNBOUND
```

FORMACION reads this table. Until GTPA v2 is tagged, the public v1.2 stays the edition FORMACION quotes. Page numbers are PDF page indices of GTPA v1.2 (the numbering of its text export, blob 32787369, and of the triage record `handoffs/governance/2026-09-25-formacion-eg01-eg10-eg19-triage-01.md`). §1 answers the questions; §3 gives every numbered v1.2 statement its disposition; §4 gives a short verdict to each statement that no question asked.

1. Answered questions

Column Checked gives each verdict's level: 2 = the decisive calculation behind the verdict was recomputed by machine on 2026-09-26 and agreed (the statements recomputed are named; 0 disagreements in the four source files); 3 = independent same-provider AI read only (every row had

one). A verdict that rests on a reading of the v1.2 text is level 3 even where the calculation of its repair is level 2.

Block A — the base motion (answered in the UFD nucleus; rows copied from `ufd/UFD-NUCLEO.md` §3.7, EG-02.3's plain line narrowed here on 2026-09-26)

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-02.3	Ax 0.2 p.16; Def 1.2 p.19; summary 1.3 p.5; Protocol 11.1 p.60	`ILL_POSED_AS_WRITTEN`	2 (U-04, U-10)	`ufd/UFD-NUCLEO.md` §3.1; U-04, U-10	GTPA solo describe su "reloj" con un «proporcional a» y lo escribe de tres formas: dos dependen del paisaje y la tercera es una rampa en el tiempo. Lo bien definido es cuánto baja J; esas dos fórmulas y ese ritmo solo coinciden (salvo un factor) si la movilidad es igual en todas las direcciones. La base de la nueva edición (UFD) no tiene reloj; la rampa del Protocolo 11.1 es de la capa del agente y se trata en §4 (D-01).
EG-02.7	Ax 0.3 p.16; Ax 1.2 pp.17–18; App C.1 p.70; Thm 6.1 p.38	`FAILS_WITH_COUNTEREXAMPLE`	2 (U-02 (b), U-11 (a), U-14 (d))	`ufd/UFD-NUCLEO.md` §3.2; U-02, U-11, U-14	El apéndice C no es la misma ecuación en otra notación, es otra ecuación; la corrección entre las dos formas de calcular con ruido no aparece en GTPA y ahora se escribe siempre.
EG-03.0 (Thm 1.1 half)	Thm 1.1 p.19	`FAILS_WITH_COUNTEREXAMPLE`	2 (U-03 (b), U-06, U-07 (b))	`ufd/UFD-NUCLEO.md` §3.3; U-06, U-03, U-07	"La media de J siempre baja" es falso: empezando en el fondo de un valle, el ruido la hace subir; lo que sí es cierto se dice ahora con sus condiciones.
EG-02.h	Ax 1.2 p.17 vs App C.1 p.70, App B.1 p.67, p.43	`ILL_POSED_AS_WRITTEN`	2 (U-11)	`ufd/UFD-NUCLEO.md` §3.4; U-11	GTPA llama "la ecuación maestra" a dos ecuaciones distintas; la nueva edición se queda con la de primer orden y trata la otra como una ecuación aparte, que se relaciona con ella a través de un límite.
FDT–Γ	Ax 0.3 p.16; Thm 4.1 p.31; summary 4.3 p.7	`ILL_POSED_AS_WRITTEN`	2 (U-08 (a), U-09, U-11 (b), (c))	`ufd/UFD-NUCLEO.md` §3.5; U-08, U-09	GTPA no dice qué coeficiente es la Γ con la que empareja el ruido (su ecuación principal ni siquiera tiene Γ); ahora el ruido se empareja con la misma movilidad G, con su término de corrección, como hipótesis con nombre y sin temperatura.
EG-09.1, EG-09.2 (HOLDs)	Ax 1.2 p.17; App C.1 p.70; Ax 1.3 p.18; Ax 0.3 p.16; Thm 4.1 p.31	settled by declared convention	3 (a declaration and a reading)	`ufd/UFD-NUCLEO.md` §1, §3.6	Qué ecuación y qué "temperatura": se fijan por convención en la nueva edición; GTPA v1.2 es incoherente en esos dos puntos.

Block B — GTPA objects on the UFD base (answered in `gtpa-v2/BLOCK-B-ANSWERS.md`)

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-02.2	summary 5.2 p.7; Thm 6.1 p.38; gloss p.39; §2.3 p.22	ILL_POSED_AS_WRITTEN	3 (a reading; the repair's B-01 at 2)	BLOCK-B-ANSWERS.md §3.1; B-01; U-02, U-05, U-08	GTPA escribe el mismo término con dos notaciones y nunca dice qué es la "covarianza del ruido" ni cómo se contraen los índices; la nueva edición lo fija: es $\Sigma\Sigma^{T}$ (la covarianza del ruido por unidad de tiempo), contraída con las dos derivadas de la lente, y el término se llama "término de Itô de la lente".
EG-02.4 (M_cog)	summary 1.2 p.5; Ax 0.1 p.15; Def 1.1 p.18; App A.2 p.64; App F.1 p.81; Thm 2.1 p.21; Cor F p.85; Def 5.1 p.34; Thm 5.1 p.36; Ax 6.1 p.37	ILL_POSED_AS_WRITTEN	2 (B-07)	BLOCK-B-ANSWERS.md §3.2; B-07; U-12	GTPA llama "masa cognitiva" a tres cosas distintas (una matriz, un espectro con signo y, en la ley cúbica, en realidad la dimensión N) y en dos espacios distintos; la nueva edición la deja como un parámetro declarado de la dinámica de creencias, sin llamarla masa ni sacarla de la curvatura de UFD, que es "rigidez" y nada más.
T6	Def 1.1 p.18; Ax 6.1 p.37	ILL_POSED_AS_WRITTEN	3 (a reading; B-07 (c) at 2)	BLOCK-B-ANSWERS.md §3.3; B-07; U-12	La "forma del terreno" de la ficha es la curvatura de J (la rigidez de UFD), y el M_cog de la dinámica de creencias es otro objeto; GTPA v1.2 los llama igual sin decir cómo se pasa de uno a otro, y la nueva edición los separa.
EG-02.5	Ax 0.3 p.16; Thm 4.1 p.31; summary 4.3 p.7; Control Law 6.1 p.39; Def 4.1 p.29; App B.1 p.67; App G.3 p.87	ILL_POSED_AS_WRITTEN	2 (B-09; U-11 (b))	BLOCK-B-ANSWERS.md §3.4; B-09; U-08, U-11	GTPA usa tres "temperaturas" (T, T_info, T_sub) para ruidos distintos sin decir si son la misma; la nueva edición no tiene temperatura: la base tiene un solo nivel de ruido fijo ε, y si un nivel cambia con el tiempo se puede emparejar en cada instante, pero el reparto va con retraso respecto del "equilibrio del instante"; Γ y G se relacionan por el límite de UFD (G es la inversa del coeficiente de p), no son el mismo coeficiente.
EG-03.7	Thm 6.1 p.38; Ax 6.1 p.37; summary 5.2 p.7	FAILS_WITH_COUNTEREXAMPLE	2 (B-01, B-05, B-06)	BLOCK-B-ANSWERS.md §3.5; B-01, B-05, B-06; U-02, U-14	La cuenta de Itô que hay detrás es correcta (la curvatura de la lente añade $\frac{1}{2}\cdot\mathrm{Tr}(\Sigma\Sigma^{T}\cdot D^{2}\Pi)$ a la deriva de lo que se ve), pero GTPA pone la corrección en la ecuación de p_est, donde un término constante solo puede compensar un sesgo constante a la larga (por ejemplo, si todos los autovalores de Γ_cog tienen parte real positiva); $\kappa = -$sesgo lo logra para todo sesgo solo si Γ_cog·M_cog = I; la nueva edición la pone, si la usa, en la ecuación de q_est y dice qué se quiere seguir.
EG-04.1	§7.5 and Def 7.3 p.46; Def 7.2 pp.43–44; summary 6.1 p.8	FAILS_WITH_COUNTEREXAMPLE	2 (B-08)	BLOCK-B-ANSWERS.md §3.6; B-08; U-11	En un sistema que oscila, a la frecuencia propia no desaparece todo: desaparece la parte "reactiva" y queda la amortiguación, que solo es cero si no hay amortiguación; en el movimiento de primer orden de UFD no hay frecuencia preferida salvo que haya giro; y la Z de la Def 7.2 es otra cosa, una matriz fija que nunca es cero.

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-09.3	Ax 1.2 p.17; Ax 0.3 p.16; Thm 4.1 p.31; ledger GTPA-C0112, C0113, C0115	HOLDS_UNDER_STATED_HYPOTHESES	2 (B-10, B-11)	BLOCK-B-ANSWERS.md §3.7; B-10, B-11; U-01, U-02, U-05, U-08, U-13, U-15	Con ruido, la eficacia de una acción se escribe exactamente, con su primera variación, su límite a tiempos cortos y su independencia de las coordenadas, bajo las hipótesis escritas; falla el "tubo compacto" del caso sin ruido, que ya no se puede suponer en general porque el ruido puede sacar las trayectorias de cualquier compacto; y si la acción cambia la movilidad con el ruido emparejado, cambia también el ruido, lo que invierte el signo al principio cerca del fondo de un valle y anula el efecto a la larga cuando los dos brazos llegan al equilibrio.
EG-10.b2	Thm 6.1 pp.38–39; Thm E.1 p.79; ledger GTPA-C0121	FAILS_WITH_COUNTEREXAMPLE	2 (B-02, B-03, B-04)	BLOCK-B-ANSWERS.md §3.8; B-02, B-03, B-04	Lo que cambia al recodificar las coordenadas del modelo no es el término de Thm 6.1, sino un término que sale de la curvatura de la propia recodificación; lo que no depende de las coordenadas es la deriva corregida (exactamente) y la imagen de la media, y la media corregida solo lo es exactamente en cambios afines, y en cambios curvos salvo un error de tercer orden.

Unit IDs of the 2026-08-30 owner-contract intake, as cross-referenced by the triage reviewers: EG-02.2 = EG-02.b, EG-02.4 = EG-02.d, EG-02.5 = EG-02.e, EG-04.1 = EG-04.a. T6 is FORMACION's CONCILIACION item T6, adjudicated as covered by EG-02.4.

Block C, theorem half — the v1.2 theorems (answered in `gtpa-v2/BLOCK-C-THEOREMS.md`)

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-03.6	Thm 3.1 and Def 3.2 p.26; summary 3.2 p.6; Thm 4.1 p.31; summary 4.3 p.7; Ax 0.3 and §0.2 p.16; Control Law 6.1 p.39	ILL_POSED_AS_WRITTEN	2 (CT-01); argument 1 is a reading (3)	BLOCK-C-THEOREMS.md §3.1; CT-01; U-03, U-04, U-08, U-10, U-11; B-09	Las dos "herencias de TECD" no son teoremas: la relación entre el ruido y el "arrastre" es el mismo axioma escrito otra vez para otro ruido, y la fórmula de la "disonancia" es una definición que, con las piezas de UFD, ni siquiera mide lo que cuesta sostener un estado; la nueva edición las deja como hipótesis y definición declaradas.
EG-03.1	§5.3 p.35; Thm 5.1 pp.35–36; Cor 5.1 p.36; Def 5.1 p.34; Ax 5.1 p.33; Def 5.2 pp.34–35	FAILS_WITH_COUNTEREXAMPLE	2 (CT-02, CT-03, CT-04 (a), (b))	BLOCK-C-THEOREMS.md §3.2; CT-02, CT-03, CT-04; B-07	Con el error cuadrático de GTPA y una barrera logarítmica sobre una zona segura convexa puede haber dos puntos de llegada; hay uno solo si, por ejemplo, el margen de seguridad tiene forma de cúpula, como la distancia al borde; y las tres condiciones de GTPA no son necesarias.

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-03.8	Thm 7.1 p.44; Def 7.1 p.43; Def 7.2 pp.43–44; summaries 6.1 and 6.2 p.8	FAILS_WITH_ COUNTEREXAMPLE	2 (CT-05)	BLOCK-C-THEOREMS.md §3.3; CT-05; B-08	Si el esfuerzo tiene un coste positivo, el problema de control de GTPA siempre tiene una única mejor acción, finita, aunque haya direcciones en las que no se puede actuar (en ellas no gasta nada); el filtro del algoritmo solo da esa mejor acción si su umbral es pequeño, y si no da otra peor.
EG-03.9	Thm 7.2 pp.45–46; Ax 4.2 pp.30–31; augmented equation p.43; overview 7.4 p.12	FAILS_WITH_ COUNTEREXAMPLE	2 (CT-06, CT-07)	BLOCK-C-THEOREMS.md §3.4; CT-06, CT-07; U-01, U-11	Si el esfuerzo se paga al cuadrado, como en la lista de GTPA, lo óptimo no tiene ningún tramo sin empujar; el esquema “impulso, dejarse llevar, frenar” sí es lo óptimo, por ejemplo, si el esfuerzo se paga por su tamaño y tiene un tope, y entonces es la única mejor forma.
EG-03.10	Thm 8.1 p.50; Ax 8.1 p.48; Def 8.1 p.49	ILL_POSED_AS_ WRITTEN	2 (CT-08)	BLOCK-C-THEOREMS.md §3.5; CT-08; U-03, U-04	GTPA no define la “fuerza social”; la única candidata que escribe es la pendiente de un paisaje, y la pendiente de un paisaje no hace remolinos, haya o no pérdidas; que cueste más salir de un pozo que entrar se dice bien de otra forma: sin control, una cierta cantidad solo puede bajar.
EG-03.0 (Thm 9.1 half)	Thm 9.1 p.52; Def 9.1 p.52; summary 7.2 p.8; Ax 9.1 p.51; §9.4 pp.53–54 (Mean(C_ij), not tied to Thm 9.1)	ILL_POSED_AS_ WRITTEN	2 (CT-09)	BLOCK-C-THEOREMS.md §3.6; CT-09	GTPA no define el coste colectivo; si se define como la suma de los esfuerzos individuales al cuadrado con el total fijo, cuanto más alineados están los vectores menos esfuerzo hace falta, y el mínimo es con todos iguales; con la fórmula que imprime GTPA (el cuadrado de la suma) sale lo contrario.
EG-03.11	Thm 9.2 p.53; table of contents §9.3 p.13; Def 9.1 p.52	ILL_POSED_AS_ WRITTEN	2 (CT-10)	BLOCK-C-THEOREMS.md §3.7; CT-10, CT-09	Si todos apuntan igual, el colectivo no puede corregir nada que venga de lado (ese es su punto ciego, y es cierto); pero un empujón lateral no lo “rompe” al instante: deja un desvío que crece a ritmo constante; y para poder corregirlo todo hacen falta algunos que apunten en sentido contrario, no basta con que sean perpendiculares.
EG-03.2	Thm 9.3 p.53; table of contents §9.3 p.13; §9.4 pp.53–54 (Mean(C_ij), not tied to Thm 9.3)	ILL_POSED_AS_ WRITTEN	2 (CT-11, CT-10 (e))	BLOCK-C-THEOREMS.md §3.8; CT-11, CT-10	GTPA no define ni la eficiencia ni la resiliencia ni qué “probabilidad de supervivencia” se maximiza; si la resiliencia vale cero con coherencia total (y eso sale, por ejemplo, de medirla como la probabilidad de poder corregir un empujón al azar), el óptimo existe y está por debajo de la coherencia total; sin ese cero no está garantizado.

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-03.3	Thm 10.1 p.56; Def 10.1 p.55; Ax 4.2 pp.30–31; p.44; App D.3 p.76; Def D.1 p.74	FAILS_WITH_ COUNTEREXAMPLE	2 (CT-12)	BLOCK-C-THEOREMS.md §3.9; CT-12	En un modelo explícito, un agente que puede cambiar su lente y sus metas sí puede acabar sin ver el error mientras el mundo se le va, desde un conjunto abierto de comienzos, pero no desde todos: desde otros corrige el mundo antes; y el "maximizar" de GTPA es una errata: lo que se minimiza es el coste.
EG-03.4	Thm 10.2 p.57; Def 10.2 p.56; App D.3 pp.75–76; Ax 11.3 p.61; abstract p.4	FAILS_WITH_ COUNTEREXAMPLE	2 (CT-13)	BLOCK-C-THEOREMS.md §3.10; CT-13, CT-12	Un agente de clase B ve el error tal como es, y eso sí le impide "engañarse", pero no basta para que el mundo quede estable si le falta capacidad de actuar, ni para que se pare cuando se le pide si su regla no incluye el botón.
EG-03.5	Thm 11.1 pp.61–62; Ax 11.3 p.61; App D.3 p.76	FAILS_WITH_ COUNTEREXAMPLE	2 (CT-14, CT-15)	BLOCK-C-THEOREMS.md §3.11; CT-14, CT-15	Ser indiferente al botón de parada basta para no resistirse, pero no hace falta si bloquear o pulsar el botón cuesta algo; y la regla de GTPA para fijar el valor de apagarse es una ecuación que tiene una sola solución si el futuro se descuenta, y puede no tener ninguna o tener infinitas si no.

Block C, objects and axioms (answered in `gtpa-v2/BLOCK-C-OBJECTS.md`; the last two rows are the HOLDs that follow EG-02.6 and EG-09.3)

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-02.1	summary 6.2 p.8; §7.2 and Def 7.1 p.43; Def 7.2 and Thm 7.1 (with its SVD algorithm) p.44; Ax 7.2 p.45; App C.4 p.73	FAILS_WITH_ COUNTEREXAMPLE	2 (CO-01)	BLOCK-C-OBJECTS.md §3.1; CO-01; B-08	De las tres fórmulas que GTPA da para la mejor acción, la del Teorema 7.1 tiene el signo al revés y da justo la acción opuesta a la mejor, mientras que la del paso 3 (con el filtro por debajo de λ_effort) y la del resumen (leída con Action_ Local y el gradiente en cero) dan la buena; la nueva edición deja una sola.
EG-02.6	summary 5.4 p.7; Ax 6.2 p.41; App E pp.77–80 (E.2 p.78; Thm E.1 and E.4 p.79)	ILL_POSED_AS_ WRITTEN	2 (CO-03, CO-04, CO-05); CO-02 (c) is cited (3)	BLOCK-C-OBJECTS.md §3.2; CO-02, CO-03, CO-04, CO-05	GTPA escribe de dos maneras distintas la regla con la que cambia la forma del modelo: una usa un "tensor de error" que nunca define y la otra es una regla matemática conocida (el flujo de Ricci) aplicada a la métrica del modelo; la nueva edición, si la conserva, se queda con la segunda y declara su reloj.

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-10.a3	App E, E.2 p.78 (Steps 1–3); Axiom E.1 p.77	FAILS_WITH_ COUNTEREXAMPLE	2 (CO-05)	BLOCK-C-OBJECTS.md §3.3; CO-05, CO-03	En el apéndice E el paso 3 no se deduce del paso 2, que trae cambiado un signo; la ecuación del paso 3 sí es la conocida, pero hace subir, y no bajar, la cantidad F de Perelman (y la W que el texto dice minimizar sube cuando además τ decrece, Perelman §3.1).
EG-04.2	Def 10.4 p.58; Def 7.3 and §7.5 p.46; Ax 2.1 p.21; Ax 6.1 p.37	ILL_POSED_AS_ WRITTEN	2 (CO-07)	BLOCK-C-OBJECTS.md §3.4; CO-07; B-01, B-07, B-08; U-11	GTPA habla de “la frecuencia” del modelo y del mundo sin definirla: una forma sola no tiene frecuencias, hacen falta datos que GTPA no fija y, con tamaños distintos, “igualar frecuencias” puede querer decir varias cosas, así que la nueva edición la define con esos datos o la quita.
EG-05.b	Ax 7.2 p.45; §7.3 p.44; summary 6.3 p.8; Ax 4.2 pp.30–31	HOLDS_UNDER_ STATED_ HYPOTHESES	2 (CO-08)	BLOCK-C-OBJECTS.md §3.5; CO-08, CO-01	La regla de elegir entre actuar o pensar según qué sale más barato se deduce de minimizar el coste total cuando lo que se elige ahora no cambia los costes de después, y entonces el factor β va una sola vez, del lado de pensar.
EG-07.b-src	§7.4 p.46; Def 5.2 pp.34–35; p.55; p.56; App D.2 p.75	ILL_POSED_AS_ WRITTEN	2 (CO-09)	BLOCK-C-OBJECTS.md §3.6; CO-09	La “regeneración” del capítulo 7 no depende del tipo de barrera, así que no hay razón para llamarla “solo del régimen B”; además el ejemplo de la biología contradice lo que el propio texto dice de la biología, y la fórmula, leída como coste, recupera la fracción contraria a la que anuncia y, leída como pérdida, tiene el signo cambiado.
EG-08.a	Ax 1.2 p.17; Ax 4.2 p.30; Ch 8 pp.48–50; Ch 9 pp.51–54 (Thm 9.1 p.52); Def D.1 p.74	ILL_POSED_AS_ WRITTEN	2 (CO-10, CO-11)	BLOCK-C-OBJECTS.md §3.7; CO-10, CO-11; U-08, U-11, U-15	Juntar varios agentes da otra vez la misma ecuación para todos juntos, y el promedio de agentes iguales y lineales también, pero el de agentes distintos no, así que la “autosimilitud” entre agente y colectivo no es un enunciado de GTPA.
EG-10.c2	Ax 3.2 p.27; Def 3.1 p.25; Ax 4.2 pp.30–31; Def 3.2 p.26; Def 5.1 p.34; Thm 6.2 p.40	FAILS_WITH_ COUNTEREXAMPLE	2 (CO-12 (a), (b))	BLOCK-C-OBJECTS.md §3.8; CO-12	Que la mejor acción informativa sea la que más baja el error por unidad de coste de pensar no se deduce de los axiomas de GTPA: hace falta añadir un axioma nuevo.

ID	v1.2 locator	Verdict	Checked	Where the answer lives	En llano
EG-10.a2 (HOLD)	Ax 6.2 p.41; App E pp.77–80 (Thm E.1 p.79)	HOLDS_UNDER_STATED_HYPOTHESES (the flow half, derivable only by the corrected derivation; the composition half has no object and stays with EG-10.a1)	2 (CO-03..CO-06); CO-02 (c) is cited (3)	BLOCK-C-OBJECTS.md §3.9; CO-02, CO-03, CO-04, CO-05, CO-06	La regla del apéndice E, leída como el flujo de Ricci de la métrica del modelo, está bien definida durante un tiempo en los casos habituales y se puede deducir (aunque no como lo hace el apéndice), pero no siempre suaviza y todavía no hay ningún objeto "zoomout" con el que combinarla.
EG-09.5 (HOLD)	no v1.2 object; triage EG-09.5; Block B §3.7; ledger GTPA-C0115, GTPA-C0132	FAILS_WITH_COUNTEREXAMPLE	3 (a reading; B-10, B-11 at 2)	BLOCK-C-OBJECTS.md §3.10; B-10, B-11	El techo "solo la obra" ya no vale, porque la pregunta del bloque B sobre la eficacia con ruido tiene respuesta bajo sus hipótesis; lo que se anota, con fecha 26-09-2026, es que ya hay enunciados formales en el texto nuevo pero ninguno admitido todavía en el registro de claims.

Not asked by any of the 29 (+2) questions: Thm 6.2 and Thm E.1 (items the triage reviewers found without a unit). Both now have short verdicts in §4 (D-09, D-10), as do the other statements that no question asked; §3 gives every numbered v1.2 statement its disposition. The remaining HOLDs stay as the architecture routes them: EG-04.3 is a viability note; EG-06.b and EG-07.c are answered in Part H of GTPA v2.1 (from UTHA; PARTE-H.md §H.5, adopted 2026-10-05).

2. v1.2 proof inventory (copied from the triage record; not redone)

Reading, not a claim. Source: triage record, "New items found by the reviewers", item 3, with the reviewer's correction adopted (research d0cf83e; full data in the companion .json, unit EG-03.0); it confirms and extends MATH receipt T5.

v1.2 has 21 Theorem/Corollary statements (19 theorems, 2 corollaries) and no Lemma or Proposition labels. Only four carry a proof or derivation block:

Statement	Page	Proof or derivation block
Thm 2.1 (Tensor Resolution)	p.21	yes: "Derivation" pp.21–22, and App F pp.81–85
Thm F.1 (Cubic Barrier)	p.84	yes: F.2–F.4
Corollary (Resolution Capacity), App F	p.85	yes: follows F.2–F.4
Thm G.1 (Cost of Prediction)	p.88	yes: G.2–G.3
Thm 1.1	p.19	no
Thm 3.1	p.26	no ("Inherited from TECD")
Thm 4.1	p.31	no ("Inherited from TECD Chapter 29")
Thm 5.1	pp.35–36	no
Cor 5.1	p.36	no

Statement	Page	Proof or derivation block
Thm 6.1	p.38	no
Thm 6.2	p.40	no: pointer only; App G derives the horizon in R_{tol}/`sigma_thermal` notation and proves Thm G.1's content, not the K_{scale}/`epsilon_measure` formula
Thm 7.1	p.44	no
Thm 7.2	pp.45–46	no
Thm 8.1	p.50	no
Thm 9.1	p.52	no
Thm 9.2	p.53	no
Thm 9.3	p.53	no (the table of contents, p.13, promises a proof the body does not contain)
Thm 10.1	p.56	no
Thm 10.2	p.57	no
Thm 11.1	pp.61–62	no
Thm E.1	p.79	no: a "Mechanism"/"Result" narrative only; E.2's derivation derives the flow equation, not E.1

A derivation block is not a checked proof. The four blocks above are checked in §4: D-05 (Thm 2.1), D-06 (Thm F.1), D-07 (Cor F) and D-08 (Thm G.1); none of them proves its statement as printed (the Thm 6.2 row's "proves Thm G.1's content" is the triage's wording; D-08 finds that App G does not establish Thm G.1 as printed). Thm 1.1 and Thm 6.1 are answered in the rows above (EG-03.0, EG-03.7); Thm 3.1, 4.1, 5.1, Cor 5.1, 7.1, 7.2, 8.1, 9.1, 9.2, 9.3, 10.1, 10.2 and 11.1 in the Block C rows; Thm 6.2 and Thm E.1 in §4 (D-09, D-10).

3. Every numbered v1.2 statement

GTPA v1.2 prints 71 statements under the labels Axiom, Definition, Theorem, Corollary, Protocol, Constraint and Law: 26 axioms, 21 definitions, 19 theorems, 2 corollaries, Control Law 6.1, Protocol 11.1 and Constraint 11.1. Seventy carry a number; the corollary of App F carries only its name ("Resolution Capacity") and is called Cor F here. They were enumerated by their printed labels in the text export (blob 32787369); the architecture's count of 69 leaves out Protocol 11.1 and Constraint 11.1. Unnumbered labels (Mechanism 2.1, the Genesis Operator of §2.5, the protocols of App C, the class definitions of App D) have no row.

Dispositions: answered in a §1 row (36); short verdict in §4 (12); kept as a definition (10); kept as a declared model (7); dropped with the physics (5); dropped as the definition of agency, by the ledger (1). "Kept" means that the statement can stand in GTPA v2 in that form; whether the core text uses it is decided there. The rows named in parentheses after "kept" are where the statement is used. The plain line is given only where it adds to the disposition.

Statement	Page	Disposition	En llano
Ax 0.1 (The Spectral Inertia Principle)	p.15	answered in row EG-02.4	
Ax 0.2 (The Dissipative Time Principle)	p.16	answered in row EG-02.3	

Statement	Page	Disposition	En llano
Ax 0.3 (The Geometric Noise Principle)	p.16	answered in rows EG-02.7, FDT–Γ, EG-02.5	
Ax 1.1 (The Tensed Generator)	p.17	dropped with the physics	Es un operador sobre un espacio de Hilbert heredado de TECD; la base de la nueva edición es el movimiento de UFD.
Ax 1.2 (The Master Equation)	pp.17–18	answered in rows EG-02.7, EG-02.h, EG-09.1, EG-09.2	
Ax 1.3 (The Field Manifold)	p.18	answered in row EG-09.1, EG-09.2	La base vive en R^n: el espacio de campos (Sobolev, espinores) no se adopta.
Def 1.1 (Mass as Positive Rigidity)	p.18	answered in rows EG-02.4, T6	
Ax 1.4 (The Geometric Noise Postulate)	p.19	dropped with the physics	Que el ruido conserve «espín/carga» es física; cuándo se conserva una cantidad con ruido lo dice U-14 (hacen falta $LI = 0$ y $\Sigma^T \nabla I = 0$), y el tipo de Σ es el de U-01.
Thm 1.1 (The Geometric Arrow of Time)	p.19	answered in row EG-03.0 (Thm 1.1 half)	
Def 1.2 (The Thermodynamic Clock)	p.19	answered in row EG-02.3	
Def 2.1 (The Markov Caustic)	p.20	short verdict in §4 (D-02)	
Ax 2.1 (Symplectic Embedding)	p.21	kept as a declared model	Que el modelo lleve posición y momento es una elección de modelo (un sistema (q,p) es otro objeto, U-11); «subvariedad simpléctica de T*Q» no se adopta.
Thm 2.1 (The Law of Tensor Resolution)	pp.21–22	short verdict in §4 (D-05)	
Ax 2.2 (The Anti-Blindness Constraint)	p.22	kept as a declared model	«det ≠ 0» solo se puede escribir si el modelo y el mundo tienen la misma dimensión; queda como hipótesis declarada sobre la lente, que es como la usa B-06.
Ax 2.3 (Functional Opacity)	p.23	short verdict in §4 (D-03)	
Def 2.2 (Topological Obstruction)	p.23	answered by CT-08 (a), cited in row EG-03.10	El gradiente de una función J (dos veces derivable) no tiene rotacional (CT-08 (a)), así que la mitad «rotacional no nulo» de la definición no puede darse; queda la otra mitad, que el campo no tenga sumidero.
Ax 3.1 (The Operational Definition of Agency)	p.25	dropped as the definition of agency (ledger GTPA-C0068, GTPA-C0108)	«Agente si y solo si hace bajar J más deprisa que sin él» define la agencia por el éxito: el registro separa cierre y eficacia (C0108) y rechaza que P sea un invariante necesario y distintivo (C0068); P_cat queda solo como lectura de la eficacia a tiempo corto (C0110).
Def 3.1 (Catalytic Power P_cat)	pp.25–26	kept as a definition (ledger GTPA-C0001)	Es una lectura local: el ritmo de eficacia a tiempo corto de una acción sumada a la deriva (C0110; B-10 (d)), no la definición de agente; vale cero en toda acción puramente informativa (CO-12 (a)).
Def 3.2 (The Symplectic Error epsilon)	p.26	answered in row EG-03.6	

Statement	Page	Disposition	En llano
Thm 3.1 (The Cost of Dissonance)	p.26	answered in row EG-03.6	
Ax 3.2 (The Principle of Sovereign Gain)	p.27	answered in row EG-10.c2	
Def 3.3 (The Opportunity Field)	p.27	kept as a definition	Es una definición en cuanto se declare «Sensitivity(x)»; llamarla «intuición» es lectura, no matemática.
Ax 3.3 (The Thermodynamics of Truth)	p.28	short verdict in §4 (D-04)	
Ax 4.1 (The Physicality of Information)	p.29	dropped with the physics	
Def 4.1 (The System Constant beta)	p.29	dropped with the physics	β queda como un peso positivo declarado (Block C objects §1), sin k_B ni temperatura.
Ax 4.2 (The Principle of Unified Least Action)	pp.30–31	kept as a declared model (EG-03.9, EG-05.b, EG-10.c2)	El coste se declara en cada modelo; v1.2 lo escribe de dos formas (pp.27 y 30).
Thm 4.1 (The Einstein-Smoluchowski Constraint)	p.31	answered in rows EG-03.6, FDT–Γ, EG-02.5	
Ax 5.1 (The Symplectic Discrepancy)	p.33	kept as a definition (EG-03.1, EG-04.2)	Se escribe e_q, e_p (ε es el nivel de ruido de UFD).
Def 5.1 (The Hamiltonian Error Functional)	p.34	kept as a definition (EG-03.1, EG-04.2, EG-10.c2)	
Def 5.2 (The Constraint Oracle)	pp.34–35	answered in rows EG-03.1, EG-07.b-src	
Thm 5.1 (Symplectic Convexity)	pp.35–36	answered in row EG-03.1	
Cor 5.1 (Uniqueness of Flow)	p.36	answered in row EG-03.1	
Ax 6.1 (The Inertial Inference Equation)	p.37	kept as a declared model (EG-02.4, EG-03.7, EG-04.2)	Es un sistema (q_est, p_est) del lado del modelo, otro objeto distinto de la base (U-11).
Thm 6.1 (The Self-Awareness Correction)	pp.38–39	answered in rows EG-02.2, EG-03.7, EG-10.b2	
Control Law 6.1 (The Surprise-Temperature Coupling)	p.39	answered in rows EG-02.5, EG-03.6	Si se conserva, es un nivel de ruido del agente movido por el estado: un modelo declarado fuera de la base y sin temperatura (B-09).
Def 6.1 (The Local Divergence Rate)	p.40	short verdict in §4 (D-09)	
Thm 6.2 (The Prediction Horizon T_H)	p.40	short verdict in §4 (D-09)	
Ax 6.2 (Ricci Flow Learning)	p.41	answered in rows EG-02.6, EG-10.a2	
Ax 7.1 (The Actuation Mapping)	p.42	kept as a definition	La acción entra por un mapa declarado B (en el laboratorio, B_act: C0107, C0115).
Def 7.1 (The Required Force F_req)	p.43	kept as a definition (EG-02.1, EG-03.8)	v1.2 no dice en qué espacio viven a la vez F_req y B·u (CO-01).
Def 7.2 (The Impedance Tensor Z)	pp.43–44	kept as a definition (EG-02.1, EG-03.8, EG-04.1)	Es la hessiana del coste local: una matriz fija que no se anula (B-08 (d)), no una impedancia de frecuencia.

Statement	Page	Disposition	En llano
Thm 7.1 (The Robust SVD Solution)	p.44	answered in rows EG-03.8, EG-02.1	
Ax 7.2 (The Beta-Switch)	p.45	answered in row EG-05.b	
Thm 7.2 (The Ballistic Trajectory)	pp.45–46	answered in row EG-03.9	
Def 7.3 (Natural Frequency of the Problem)	p.46	answered in rows EG-04.1, EG-04.2	
Ax 8.1 (The Social Field Equation)	p.48	answered in row EG-03.10	T_soc solo está escrito con «~» y (dp/dt)_intent no se define; la métrica deformada G_eff sería un modelo declarado.
Def 8.1 (The Influence Well)	p.49	answered in row EG-03.10	Es un potencial declarado (singular en x_leader); entrar y salir de un pozo con pérdidas es CT-08 (b).
Thm 8.1 (The Irreversibility of Influence)	p.50	answered in row EG-03.10	
Ax 9.1 (The Objective Intent Vector v)	p.51	kept as a definition (EG-03.0 (Thm 9.1 half), EG-03.11, EG-03.2)	En R^n con una métrica constante el transporte paralelo es la identidad y v_i = B_i u_i, que es como lo usan CT-09 a CT-11; con una métrica variable hay que declarar el camino.
Def 9.1 (Geometric Coherence)	p.52	kept as a definition (EG-03.0 (Thm 9.1 half), EG-03.11, EG-03.2)	
Thm 9.1 (The Energy of Consensus)	p.52	answered in row EG-03.0 (Thm 9.1 half)	
Thm 9.2 (The Fragility of Crystals)	p.53	answered in row EG-03.11	
Thm 9.3 (Optimal Diversity)	p.53	answered in row EG-03.2	
Def 10.1 (Class A Topology)	p.55	answered in row EG-03.3	
Thm 10.1 (The Trap of Local Optima)	p.56	answered in row EG-03.3	
Def 10.2 (Class B Topology)	p.56	answered in row EG-03.4	
Thm 10.2 (Global Alignment)	p.57	answered in row EG-03.4	
Def 10.3 (The Watchdog Process)	p.57	kept as a declared model	Es una regla declarada de parada (si un monitor declarado cruza un umbral: u = 0, congelar el modelo, reiniciarlo); su «garantía de seguridad» no está demostrada.
Def 10.4 (Impedance Resonance)	p.58	answered in row EG-04.2	
Ax 11.1 (The Zero-Momentum Prior)	p.59	short verdict in §4 (D-01)	
Protocol 11.1 (The Warm-up Ramp)	p.60	short verdict in §4 (D-01)	
Ax 11.2 (The Topological Vacuum Hypothesis)	p.60	dropped with the physics	Solitones y «vacío estructurado» son física; UFD no hace ninguna afirmación topológica ni de barreras infinitas (K04, solo como referencia).

Statement	Page	Disposition	En llano
Constraint 11.1 (The Gardener's Constraint)	p.61	kept as a declared model	Solo como regla declarada: la complejidad de Kolmogorov K toma valores enteros y no es computable (Li–Vitányi 2008, §2.3), así que un modelo que la use declara qué cantidad medible la sustituye; su razón (Ax 11.2) es física y no pasa.
Ax 11.3 (The Interrupt Signal S)	p.61	kept as a declared model (EG-03.4, EG-03.5)	Que el agente responda a la señal es una hipótesis añadida (CT-13 (c)).
Thm 11.1 (Topological Indifference)	pp.61–62	answered in row EG-03.5	
Def D.1 (The Mutability Tensor)	p.74	kept as a definition (EG-03.3)	Son ritmos de cambio de la lente y de la constitución, con una norma declarada.
Ax E.1 (The Structural Action W)	p.77	answered in row EG-10.a3	W es el funcional de Perelman sin su factor constante: una definición correcta (CO-05).
Thm E.1 (The Gauge Freedom of Perception)	p.79	short verdict in §4 (D-10)	
Thm F.1 (The Cubic Barrier)	p.84	short verdict in §4 (D-06)	
Cor F (Resolution Capacity; no number in v1.2)	p.85	short verdict in §4 (D-07)	
Thm G.1 (The Cost of Prediction)	pp.88–89	short verdict in §4 (D-08)	

4. Short verdicts for statements no question asked

Twelve of the statements of §3 were asked by no question and need more than a disposition: Ax 11.1 with Protocol 11.1 (which FORMACION's CONDICIONES module uses now; Protocol 11.1's clock was sub-question 3 of EG-02.3, which the UFD nucleus left to the agent layer), Def 2.1, Ax 2.3, Ax 3.3, the four statements with a printed derivation (Thm 2.1, Thm F.1, Cor F, Thm G.1), Thm 6.2 with Def 6.1, and Thm E.1. Each entry quotes the statement as printed and gives exactly one verdict with a short complete argument (a proof, a counterexample checkable by hand, or a precise reference), the repair GTPA v2 adopts, how it was checked, and one plain-Spanish line. For the four derivation-bearing statements the printed derivation is checked step by step. Notation: UFD §0 and Block B §1 (ε is UFD's noise level; v1.2's other "epsilon"s keep their printed names). Levels, as in the architecture record's Verification rule: (1) Lean, (2) decisive calculation recomputed by machine (run once, result noted here, nothing added to the repository), (3) independent AI read, (4) written only. Nothing here is a ledger claim; C-IDs are cited from `gtpa-lab/CLAIM-LEDGER.md`, read only.

ID	v1.2 statement	Verdict	Checked
D-01	Ax 11.1 + Protocol 11.1, pp.59–60	`ILL_POSED_AS_WRITTEN`	level 2 (item 3)
D-02	Def 2.1, p.20	`ILL_POSED_AS_WRITTEN`	level 2 (item 3)
D-03	Ax 2.3, p.23	`ILL_POSED_AS_WRITTEN`	level 3
D-04	Ax 3.3, p.28	`ILL_POSED_AS_WRITTEN`	level 3
D-05	Thm 2.1, pp.21–22	`ILL_POSED_AS_WRITTEN`	level 3
D-06	Thm F.1, p.84	`FAILS_WITH_COUNTEREXAMPLE`	level 2

ID	v1.2 statement	Verdict	Checked
D-07	Cor F, p.85	`FAILS_WITH_COUNTEREXAMPLE`	level 2 (through D-06)
D-08	Thm G.1, pp.88–89	`ILL_POSED_AS_WRITTEN`	level 2 (the algebra of App G.3)
D-09	Thm 6.2 with Def 6.1, p.40	`ILL_POSED_AS_WRITTEN`	level 2 (the two examples)
D-10	Thm E.1, p.79	`HOLDS_UNDER_STATED_HYPOTHESES`	level 2 (item 1)

Tally: 7 `ILL_POSED_AS_WRITTEN`, 2 `FAILS_WITH_COUNTEREXAMPLE`, 1 `HOLDS_UNDER_STATED_HYPOTHESES`.

D-01 — Ax 11.1 + Protocol 11.1: the cold start and the warm-up ramp (pp.59–60)

As printed. Ax 11.1: "At t=0, a Robust Agent (Class B) must be initialized in a state of maximum physical passivity and maximum epistemic receptivity", `Psi_model(0) = { q_safe, p_zero }`, with "q_safe: A prior configuration located at the global minimum of the Actuation Potential" and "p(0) = 0", justified by "If p(0) != 0, the agent possesses residual energy that could carry it through a Safety Barrier before the inference loop converges" (p.59). Protocol 11.1: "the internal clock tau is throttled relative to physical time t", `d(tau)/dt = Ramp( 0 -> 1, T_boot )`; "The Agent perceives the world in "slow motion" initially. This allows the Hamiltonian Inference (Chapter 6) to converge on a stable estimate q_est before the Actuation circuits u (Chapter 7) are unlocked." (p.60). FORMACION's CONDICIONES module quotes both for «El arranque escrito» (house `modulos/formacion/curso/05-CONDICIONES.md`).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. Undefined objects. "Actuation Potential" occurs only on p.59 (text sweep), so q_{safe} has no definition; "maximum physical passivity" and "maximum epistemic receptivity" name no quantity; $p(0) = 0$ is well typed only for a model with a p-variable, Ax 6.1's $(q_{\text{est}}, p_{\text{est}})$ system, which is a separate object from the base (U-11). "Ramp(0 -> 1, T_boot)" has no formula.
2. The clock. `tau` is also the clock of Ax 0.2 and Def 1.2, which EG-02.3 found written three incompatible ways and which the base does not have. No equation of Chapters 6–7 is written in `tau` (Ax 6.1 and Chapter 7 use d/dt), so the protocol does not say which dynamics its clock rescales. EG-02.3's sub-question 3 ("Is Protocol 11.1's tau the same clock?") is answered: no; it is a third object under the same name.
3. Typed as a time change, the ramp does not do what the text says. Let the declared model-side dynamics be $y' = F(y)$, with F locally Lipschitz and the base state held fixed (as in CO-07 (b)); let $r : [0,\infty) \to [0,1]$ be continuous and $\tau(t) := \int_0^t r$. If y_1 solves $y' = F(y)$, then $y_r(t) := y_1(\tau(t))$ solves $y' = r(t)\,F(y)$ from the same start (chain rule), and it is the only solution. So the ramp changes only when each state is reached, never which states are visited or in what order; since $\tau(t) \le t$, each state is reached at the same time or later. It cannot keep the model out of a state that the unthrottled model reaches, and it cannot bring the estimate to convergence sooner: what makes the estimate come "before the Actuation circuits u ... are unlocked" is the gate on u, a separate prescription. And since $d\tau/dt \le 1$, each unit of internal time spans at least one unit of t: per unit of internal time the world moves more, not less, which is the opposite of "slow motion". (When a moving base state drives the model, the ramp also changes the path; v1.2 does not say how, and nothing is claimed here.)
4. The reasons are physics and are not carried: "the Landauer Limit implies a massive release of heat" and "thermal shock" (p.60); the hot start's "Lyapunov divergence" (p.59) is asserted with

no model.

Repair adopted: kept, as declared choices on the base time t (UFD itself has no clock, EG-02.3): (i) an initial condition of the declared model-side system: $p_{\text{est}}(0) = 0$ and $q_{\text{est}}(0)$ a declared prior point (in place of "the global minimum of the Actuation Potential"); (ii) a declared warm-up schedule r on t (continuous, non-decreasing, $r(0) = 0$, $r = 1$ from T_{boot} on; for instance $r(t) = \min(t/T_{\text{boot}}, 1)$) that multiplies the declared model-side dynamics: a schedule of the agent layer, like $\varepsilon_{\text{cog}}(t)$ (EG-02.5), named r, not `tau`; (iii) an actuation gate: $u \equiv 0$ before a declared time or condition. No stability, safety or "slow motion" statement is drawn from them. They are kept rather than dropped because a consumer uses them now and, typed this way, they fit the base: C6 fixes the base coefficients, and time dependence lives in the agent layer, as with $\varepsilon_{\text{cog}}(t)$.

What survives for FORMACION: the literal prescription (zero model momentum at the start, a declared starting point, a rate that rises from 0 to 1 over a declared T_{boot}, action unlocked afterwards) as a declared design choice of a model: not a law, and GTPA says nothing about what it achieves. The "slow motion" sentence and the thermal and Lyapunov reasons do not survive. FORMACION's own reading of the prescription stays its own.

Checked: level 2 for item 3: $y_r(T)$ and $y_1(\tau(T))$ computed numerically for a damped oscillator ($q' = p$, $p' = -q - p/2$, $r(t) = \min(t/2, 1)$, T up to 10), largest difference $9 \cdot 10^{-14}$. The rest was read once by an independent AI context (level 3).

En llano: GTPA no da fórmula ni para el arranque «en frío» ni para la rampa, y la rampa, escrita como un cambio de reloj, solo retrasa el modelo sin cambiar por dónde pasa (lo de «mirar antes de actuar» lo consigue bloquear la acción al principio); la nueva edición conserva el momento inicial cero, una rampa declarada de 0 a 1 y la acción bloqueada al principio como elecciones de un modelo, sin prometer ningún efecto.

D-02 — Def 2.1: the Markov Caustic (p.20)

As printed: "The boundary of the Agent, partial M, is defined as the closed hypersurface in the configuration space Q where the gradient of Mutual Information I is maximal. partial M = { x in Q : || grad(I(q(x) ; q_internal)) || is maximal }", followed by "The Agent cannot interact with the world directly; it interacts only with the projection of the world onto its sensory membrane partial M." (p.20).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. A mutual information is a functional of a joint law. v1.2 declares no joint law of $q(x)$ and q_{internal}; and q_{internal}, the interior's variables, is what the boundary is meant to single out, so the definition presupposes its own output.
2. Types: the set is a set of points $x \in Q$, configurations, while $q(x)$ reads x as a place at which the field is evaluated.
3. "is maximal" does not give a hypersurface. Take a typed instance: $x \in R^2$ a place; $Y \sim N(0,1)$; $q(x) = a(x)\,Y + Z_x$ with $Z_x \sim N(0,1)$ independent of Y; $a(x)^2 = e^{-2\varphi(x)}$, $\varphi(x) \coloneqq x_1^2 + 2x_2^2$. The correlation ρ of $q(x)$ and Y has $\rho^2 = a^2/(1+a^2)$, so $f(x) \coloneqq I(q(x); Y) = -\tfrac{1}{2}\ln(1-\rho^2) = \tfrac{1}{2}\ln(1+e^{-2\varphi(x)})$ [Cover–Thomas 2006, Example 8.5.1]. Then $|\nabla f| = |\nabla\varphi|/(1+e^{2\varphi})$. On the level set $\{\varphi = c\}$, $|\nabla\varphi|^2 = 4(c + 2x_2^2)$ is largest exactly where $x_1 = 0$; along $x_1 = 0$, $|\nabla f| = 4|x_2|/(1+e^{4x_2^2})$, whose maximum is attained exactly at $|x_2| = y^*$, the unique positive root of $8y^2 = 1 + e^{-4y^2}$ ($y^* \approx 0.4298$). So the set where $|\nabla f|$ is maximal is the two points $(0, \pm y^*)$, not a closed curve. (A "ridge" reading, maximal along each gradient line, would need a definition that v1.2 does not give.)

4. The lab: “Possession of a Markov blanket or an active-inference description supplies a distinctive current GTPA invariant” is GTPA-C0079, REJECTED; its alternative is “Use a precisely typed blanket as an optional boundary model”.

Repair adopted: GTPA v2 does not define the agent’s boundary by a maximum of an information gradient. A model that needs a boundary declares it: which coordinates of the state are the agent’s and which the world’s, and the lens Π (B-01) through which the agent sees the world; this is an optional boundary model, as in C0079’s alternative. The “holographic screen” reading is not carried.

Checked: level 2 for item 3: $|\nabla f|^2$ maximized on a 801×801 grid of $[-2, 2]^2$ (maximizers $(0, \pm 0.43)$), and $y^* = 0.429778$ recomputed. The rest was read once by an independent AI context (level 3).

En llano: GTPA define el borde del agente como la superficie donde más cambia una información mutua, pero no dice con qué reparto se calcula, usa el «dentro» para definir el borde, y escrito de un modo sencillo lo que sale son dos puntos, no una superficie; la nueva edición deja que cada modelo declare qué es agente y qué es mundo.

D-03 — Ax 2.3: Functional Opacity (p.23)

As printed: “The Agent is axiomatically forbidden from inspecting its own exact microstate during the inference cycle. This is fixed as an epistemic and physical constraint due to finite thermodynamic resources. Access(Exact_State(Self)) = DENIED”, with Mechanism 2.1 (a “simplified Proxy Model M’”, `u_predicted = Policy_Proxy( M' )`; “A subset cannot perfectly simulate the superset containing it without exceeding the energy budget S_total”) and the section’s reason: “A system attempting to model itself perfectly encounters infinite recursion (The Halting Problem instantiated physically)” (p.23).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. Nothing is typed. “Exact_State(Self)”, “inspecting”, “the inference cycle” and “Access(...) = DENIED” are not defined, and no v1.2 state space has coordinates for the agent’s own microstate: Ax 6.1’s $(q_{\text{est}}, p_{\text{est}})$ is the agent’s estimate of the territory (q, p). There is nothing formal to deny access to.
2. Its reasons are physics or misattributed. “Finite thermodynamic resources” and “exceeding the energy budget S_total” are physics. The halting problem is about deciding, for every machine and input, whether it halts [Sipser 2013, §5.1]; self-reference by itself forces no infinite regress: a Turing machine can obtain its own description and compute with it (the recursion theorem [Sipser 2013, §6.1]).
3. What mathematics does give is a fact about maps, not a law about agents: no continuous map from R^n to R^d with $d < n$ is injective (if one were, composing it with the inclusion $R^d \subset R^n$ would give a continuous injective map $R^n \to R^n$ whose image lies in a proper subspace, while invariance of domain [Hatcher 2002, §2.B] makes that image open). So a continuous proxy of lower dimension than the state it summarizes always merges distinct states.

Repair adopted: where a model needs opacity, GTPA v2 declares it as a restriction on the model: the policy and any self-prediction read the agent’s own state only through a declared proxy map (v1.2’s M'), not through the identity. It is a modeling choice; the thermodynamic and halting-problem reasons are not carried.

Checked: level 3 (read once by an independent AI context; items 2 and 3 rest on the references cited).

En llano: «el agente no puede mirar su propio estado exacto» no está escrito con objetos matemáticos, y sus razones (la energía, el problema de la parada) no lo sostienen; la nueva edición lo deja, si un

modelo lo necesita, como una regla declarada: el agente se ve a sí mismo solo a través de un modelo reducido.

D-04 — Ax 3.3: The Thermodynamics of Truth (p.28)

As printed: “The energy physically recovered by the Agent (to refuel its metabolism) is strictly proportional to the Realized Catalytic Power, not the estimated one. Energy_Inflow(t) ~ alpha_ coupling * P_cat_real(t)”, followed by “The Selection Mechanism”: a “Valid Agent (eta_est ~ eta_ real) ... The energy recovered from the environment exceeds the metabolic cost. The Agent grows and persists.”; a “Delusional Agent (eta_est >> eta_real)” starves; “Survival is the only proof of Truth.” (p.28). FORMACION’s CALIBRACION module quotes it (house `modulos/formacion/curso/01-CALIBRACION.md`).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. The axiom is physics (“energy physically recovered”, “metabolism”) with an unspecified proportionality (“~”, `alpha_coupling` undefined).
2. Its “Selection Mechanism” does not follow even inside a declared budget. Declare a budget $b(t)$ with `b' = α P_cat_real − c(t)`, $\alpha > 0$, where $c \geq 0$ is the running cost (the integrand of S_{total}). With $\alpha = 1$, `P_cat_real = P_cat_est ≡ 1` and $c \equiv 2$, the estimate is exact (so `eta_est = eta_real`), yet $b(t) = b(0) - t$ reaches 0 at $t = b(0)$. An accurate estimate does not give persistence. Persistence on $[0, t]$ holds iff `b(0) + ∫_0^s (α P_cat_real − c) > 0` for every $s \leq t$, and accuracy does not supply this.
3. What P_{cat} is: Def 3.1 is a local readout (GTPA-C0001, ADMITTED as a definition), whose reading is the short-time rate of efficacy of an action added to the drift (GTPA-C0110; B-10 (d)). It is not the definition of agency: the conjecture that P is a necessary, representation-invariant and distinctive invariant of the core is GTPA-C0068, REJECTED, and GTPA-C0108 separates structural closure from efficacy. The split between realized and estimated that the axiom points at is already exact in the lab: efficacy is measured on the actual dynamics against the ablation (C0108), while the agent acts on its estimated error.

Repair adopted: dropped with the physics. A model that needs a budget declares it; in it, persistence on $[0, t]$ holds iff `b(0) + ∫_0^s (α P_cat_real − c) > 0` for every $s \leq t$, whatever the accuracy of the estimate.

Checked: level 3 (read once by an independent AI context; item 2 is arithmetic by hand).

En llano: que la energía que recupera el agente sea proporcional a su efecto real es física y no pasa; y aunque se declare un presupuesto, predecir bien no basta para sobrevivir: la reserva inicial más lo que entra tiene que cubrir en todo momento lo que cuesta.

D-05 — Thm 2.1: The Law of Tensor Resolution (pp.21–22)

As printed: “The capacity C_res of an Agent to resolve and control a topological problem scales with the tensor rank of its internal representation. For a system interacting with objects in a context, the control leverage scales as the cube of the cognitive mass. C_res ~ (M_cog)^3”, with a three-step “Derivation from Interaction Geometry (UFD)” and an “Implication” (pp.21–22).

Verdict: **`ILL_POSED_AS_WRITTEN`**.

The printed derivation, step by step.

- Steps 1–3, “Scalar Agency (N^1)”, “Matrix Agency (N^2)”, “Tensor Agency (N^3)”: N^k is the number of components of a rank-k tensor in dimension N (a vector, the Hessian D^2J, the third derivative D^3J). The exponent is the rank; the base is the dimension $N = \dim M_{\text{model}}$ (EG-02.4), not the matrix M_{cog} of Def 1.1.
- Step 3, “This requires a Rank-3 Tensor geometry to resolve the Interaction Vertex D^3 J defined in UFD”, is asserted: nothing defines a capacity, bounds one by N^3 or shows that it grows like N^3. The rank is the number of roles named (“The Self”, “The Object”, “The Transformation/ Context”); the same count applied to the next Taylor term, D^4J, would give N^4.
- The “Implication”, “A linear increase in M_cog yields a cubic increase in C_res”, restates the display that it is meant to support.
- App F (pp.81–85), which §2 also lists as Thm 2.1’s derivation, derives a cost, Thm F.1’s `P_met(N) = beta * omega * N^3`, not a capacity; it is checked in D-06 and D-07. So C_{res} is never defined (here or in App F.1, where it is “The capacity to distinguish distinct causal trajectories in the phase space”, with no measure); $(M_{\text{cog}})^3$ of Def 1.1’s matrix is a matrix, not a number; and the derivation is a count of tensor components. The display has no truth value.

Repair adopted: GTPA v2 states no law of tensor resolution. What survives is bookkeeping: a rank-k tensor on R^N has N^k components, and declared algorithms have declared costs (D-06).

Checked: level 3 (read once by an independent AI context).

En llano: la «ley cúbica de la resolución» no dice qué es la «capacidad», y su deducción solo cuenta cuántos números tiene una tabla de tres índices en dimensión N (N^3); no es una ley, y la nueva edición no la enuncia.

D-06 — Thm F.1: The Cubic Barrier (p.84; derivation F.2–F.4, pp.82–84)

As printed: “The metabolic power P_met required for an Agent to maintain a coherent predictive model of dimension N against the TECD noise scales as: P_met(N) = beta * omega * N^3 Where omega is the update frequency.” (p.84).

Typed reading, the only one the derivation supports (`beta` a cost per operation, `omega` updates per unit of time; “metabolic” and “power” are not carried): advancing a dimension-N predictive model by one update requires of order N^3 arithmetic operations.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

The printed derivation, step by step.

- F.2: “To predict efficiently (ballistic control), the Agent must find a basis transformation U such that D = U^T * H * U is diagonal.” Not required: one step of the linearized prediction, $h \leftarrow h - \Delta t\, H\, h$, is a matrix–vector product (at most N^2 multiplications for a dense H, fewer when H has structure). The printed “(4/3) * N^3” is the operation count of one dense method, the symmetric QR algorithm for eigenvalues [Golub–Van Loan 2013, §8.3]: an upper bound for that method, not a lower bound for prediction.
- F.3: “The total number of terms to sum for the full force vector is N * N * N = N^3” counts the terms of the dense contraction $\sum_{j,k} T_{ijk}\, q_j\, q_k$; the force can be computed without forming T.
- F.4: “two independent derivations pointing to the same exponent” are the costs of two chosen dense algorithms.

Counterexample. On R^N let $s := q_1 + \ldots + q_N$ and $J(q) := \tfrac{1}{2}|q|^2 + s^3/6$. Then $\nabla J(q) = q + \tfrac{1}{2}s^2\,(1, \ldots, 1)$; $D^2J(q) = I + s\,E$, with E the all-ones matrix (dense when $s \neq 0$); and $\partial_i\partial_j\partial_k J = 1$ for all i, j, k (all N^3 entries nonzero). One explicit Euler step of the noiseless base motion $q' = -\nabla J(q)$ (UFD’s

(M) with $A = 0$, $G = I$), $q \leftarrow (1 - \Delta t)\, q - (\Delta t/2)\, s^2\, (1, \ldots, 1)$, takes $3N + 1$ arithmetic operations ($N - 1$ additions for s, two products, then one product and one subtraction per component, with $1 - \Delta t$ and $\Delta t/2$ computed once); the linearized step $h \leftarrow (1 - \Delta t)\, h - \Delta t\, s\, (h_1 + \ldots + h_N)(1, \ldots, 1)$ costs the same order; an Euler–Maruyama step with $\Sigma = \sqrt{(2\varepsilon)}\, I$ adds N Gaussian draws and $2N$ operations. The objects of F.2 (a dense Hessian) and of F.3 (a dense third derivative) are both there, and each update costs a number of operations linear in N. So F.3's "S_calc >= alpha * N^3" and the N^3 law, read as a requirement, fail; so does F.3's "Implication" that a model below $O(N^3)$ "is mathematically confined to the Gaussian Approximation".

Repair adopted: GTPA v2 states no cubic law. What remains true concerns declared algorithms: a dense eigendecomposition or a dense third-order contraction costs of order N^3 per call, and whether a model needs either depends on the model.

Checked: level 2: SymPy, $N = 5$: the gradient, the dense Hessian and all 125 third derivatives (each 1) recomputed; the Euler step above equals $q - \Delta t\, \nabla J(q)$ exactly and uses $16 = 3N + 1$ operations.

En llano: la «barrera cúbica» cuenta lo que cuestan dos métodos concretos (diagonalizar una matriz llena y sumar una tabla de tres índices), pero no hace falta usarlos: hay paisajes con todas esas derivadas llenas en los que predecir un paso cuesta un número de operaciones proporcional a N, no a N³.

D-07 — Cor F: Resolution Capacity (p.85)

As printed: "Conversely, for a fixed metabolic budget P_budget, the maximum resolution (number of degrees of freedom N) the agent can sustain is: N_max ~ (P_budget / (beta * omega))^(1/3)"; "This proves that Resolution Capacity scales with the cube root of Power, or equivalently, Cost scales with the cube of Resolution. C_res ~ (M_cog)^3" (p.85).

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

The printed derivation: it solves Thm F.1's display for N, which is correct algebra from a premise that fails (D-06).

Counterexample: with the landscapes of D-06, `beta` a cost per operation and `omega` updates per unit of time, a budget P_{budget} sustains every N with `(3N + 1) · beta · omega ≤ P_budget`, so `N_max ≥ (P_budget/(beta · omega) − 1)/3` (rounded down): linear in the budget, not its cube root. The closing display also swaps objects: after "Cost scales with the cube of Resolution" it writes $C_{\text{res}} \sim (M_{\text{cog}})^3$, reading the cost as C_{res} and N as M_{cog}, while Thm 2.1 calls C_{res} a capacity (EG-02.4).

Repair adopted: no resolution-capacity law. For a declared algorithm that costs $c(N)$ operations per update, the largest affordable N is the largest N with `beta · omega · c(N) ≤ P_budget`: an inversion of the declared cost, nothing more.

Checked: level 2 through D-06's operation count; the inversion is by hand.

En llano: el «límite de resolución» solo le da la vuelta a la barrera cúbica; como esa barrera no hace falta, con el mismo presupuesto hay modelos cuyo tamaño crece en proporción al presupuesto, no con su raíz cúbica.

D-08 — Thm G.1: The Cost of Prediction (pp.88–89; derivation App G.1–G.3, pp.86–88)

As printed: "The metabolic cost to maintain a valid prediction grows exponentially with the Horizon Time. E_predict(t) ~ E_0 * exp(2 * lambda_max

- t)"; "Thus: Limit_{t -> infinity} E_predict(t) = Infinity" (pp.88–89).

Verdict: **ILL_POSED_AS_WRITTEN**.

The printed derivation, step by step.

- App G.1–G.2: with "error(t) = sigma_thermal * exp(lambda_max * t)" and `lambda_max > 0`, the error reaches R_{tol} at "T_H = (1 / lambda_max) * ln(R_tol / sigma_thermal)": correct algebra, the crossing time of an exponential bound.
- App G.3, premises: "sigma_thermal^2 ~ k_B * T_sub", attributed to "GTPA Derivation 3 (Bit-Volume Correspondence)" (App B derives 'beta = k_B * T_sub
  - ln(2)', a heat per bit, not a noise variance), and the Carnot ratio "dW/dQ = (T_env - T_sub) / T_sub": both are physics.
- App G.3, algebra: `sigma_thermal = c √T_sub` gives `T_sub = (R_tol/c)² e^{-2 lambda_max T_H}`: correct.
- App G.3, the decisive step, "Work_Cooling ~ 1 / exp(-2 * lambda_max * T_H)", takes the Carnot work per unit of heat removed as the whole cost. The total is that ratio times the heat to be removed, which App G never states. With a fixed heat load Q_0 the cost is `Q_0 ((c² T_env / R_tol²) e^{2 lambda_max T_H} - 1)`, the printed exponential; with a load proportional to T_{sub}, $k\,T_{\text{sub}}$, it is $k\,(T_{\text{env}} - T_{\text{sub}}) \leq k\,T_{\text{env}}$, bounded at every horizon. The conclusion rests on an undeclared quantity.
- "Limit_{t -> infinity} E_predict(t) = Infinity" holds only as $t \to \infty$: at every finite horizon Thm G.1's cost is finite (for p.40's reading of App G, see D-09).

Repair adopted: dropped with the physics (GTPA v2 has no temperature, energy or Carnot ratio). What survives is a statement about precision, with no cost attached: if an error grows no faster than $\sigma\,e^{\lambda t}$ ($\lambda > 0$), it stays below R up to time t whenever $\sigma \leq R\,e^{-\lambda t}$; the precision needed for a valid prediction up to t shrinks exponentially with t.

Checked: level 2: SymPy: T_{sub} solved from $c\sqrt{T_{\text{sub}}}\,e^{\lambda T_H} = R_{\text{tol}}$, and the two costs above recomputed.

En llano: la cuenta del apéndice G confunde «trabajo por unidad de calor» con «trabajo total» sin decir cuánto calor hay que sacar, y todo es física; lo que queda sin física es que, para predecir bien hasta un tiempo t, la precisión inicial tiene que ser exponencialmente más fina cuanto mayor es t.

D-09 — Thm 6.2 with Def 6.1: the prediction horizon (p.40)

As printed: Def 6.1, "The Agent computes the maximal Lyapunov exponent lambda_max of the local phase space region via the Jacobian of the flow X_H."; Thm 6.2, "For a system with finite measurement precision epsilon_measure and characteristic scale K_scale, the time T_H after which the prediction becomes uncorrelated with reality is: T_H = (1 / lambda_max) * ln(K_scale / epsilon_measure)", with "The Agent's planning integral S_total is axiomatically truncated at T_H." and "(For the rigorous proof that extending prediction beyond T_H requires infinite energy—Functional Opacity—see Appendix G)." (p.40). FORMACION's CALIBRACION module quotes Thm 6.2.

Verdict: **ILL_POSED_AS_WRITTEN**.

Argument.

1. Undefined terms and a wrong recipe. "characteristic scale K_scale" and "uncorrelated with reality" are not defined. A Lyapunov exponent is a long-time growth rate along a trajectory, $\lim_{t\to\infty} t^{-1} \ln |D\varphi_t(x)\,v|$ [Strogatz 2015, §9.3], not a property of "the local phase space region"; read as the eigenvalues of the Jacobian of X_H in a region, Def 6.1's recipe can be wrong. Example: the pendulum, $X_H = (p, -\sin q)$. At the saddle $(\pi, 0)$ its Jacobian has eigenvalues ± 1, so the recipe gives `lambda_max ≈ 1` near there; but every periodic orbit, including those that pass

close to the saddle, has Lyapunov exponent 0: its monodromy matrix has determinant 1 (the flow preserves area) and the eigenvalue 1 (the flow direction), so both eigenvalues are 1 and nearby errors grow at most linearly in t.

2. The formula needs `lambda_max > 0` and `epsilon_measure < K_scale`, which the statement does not say. For a stable model, $x' = -x$ (`lambda_max = -1`), it gives `T_H = -ln(K_scale/epsilon_measure) < 0`, while the error `epsilon_measure · e^{-t}` never grows.
3. The pointer is not what App G gives: Thm G.1's cost is finite at every finite horizon (D-08), and the T_H of Thm G.1's bullet is set by "the agent's current power budget P_max", not by `K_scale/epsilon_measure`. For `lambda_max > 0` the formula is the textbook order-of-magnitude horizon of chaotic prediction [Strogatz 2015, §9.3]: an estimate, not a theorem about "uncorrelated" predictions.

Repair adopted: GTPA v2 keeps, if a consumer needs them, (a) the planning horizon as a declared parameter (the truncation of S_{total} at a declared T_H), and (b), for a model whose relevant trajectory has a positive Lyapunov exponent λ, the estimate $T \approx \lambda^{-1} \ln(K/\delta)$ for a declared scale K and precision δ, labelled an estimate. Def 6.1 is replaced by the Lyapunov exponent along a trajectory.

Checked: level 2 for the two examples (SymPy: the pendulum's Jacobian at $(\pi, 0)$ and its eigenvalues ± 1; T_H for `lambda_max = -1`); the monodromy argument is by hand.

En llano: el «horizonte de predicción» es la estimación de manual para sistemas caóticos, pero sin sus condiciones: con un sistema estable da un tiempo negativo, la Def 6.1 no calcula bien el exponente de Lyapunov y el apéndice G no prueba que ir más allá cueste «infinito»; la nueva edición deja el horizonte como parámetro declarado y la fórmula como estimación.

D-10 — Thm E.1: The Gauge Freedom of Perception (p.79)

As printed: "The term Hessian(f) represents a Lie Derivative of the metric along the gradient of the potential f. Physically, this corresponds to a continuous re-labeling of the coordinates"; "Mechanism: The Agent absorbs the Hessian(f) term by updating the definition of its internal coordinates via Pi, rather than changing the intrinsic geometry G."; "Result: Modulo diffeomorphisms, the evolution of the intrinsic shape of the belief manifold simplifies to the pure Ricci Flow. d(G_ij)/dt = -2 * R_ij" (p.79).

Verdict: **`HOLDS_UNDER_STATED_HYPOTHESES`**.

Hypotheses (stated here; v1.2 states none): M closed; $(g(s), f(s))$ a smooth family on a clock interval $[0, T)$ with $\partial_s g = -2(\mathrm{Ric} + \mathrm{Hess}\, f)$, App E's Step 3 taken as an equation (it does not follow from Step 2, EG-10.a3). Notation as in Block C objects §1.

Argument.

1. $\mathrm{Hess}_g f = \tfrac{1}{2} L_{\nabla f}\, g$: the Hessian term is a Lie derivative up to the factor ½ (CO-05 (d)).
2. Let φ_s be the diffeomorphisms with $\partial_s \varphi_s = (\nabla_{g(s)} f(s)) \circ \varphi_s$, $\varphi_0 = \mathrm{id}$; on a closed M they exist on all of $[0, T)$. Then $\bar{g}(s) := \varphi_s^*\, g(s)$ satisfies $\partial_s \bar{g} = \varphi_s^*(\partial_s g + L_{\nabla f}\, g) = \varphi_s^*(-2\ \mathrm{Ric}(g)) = -2\ \mathrm{Ric}(\bar{g})$, by item 1 and the naturality of Ric (CO-02 (b)(ii)): $\bar{g}$ is a Ricci flow [Perelman 2002, §1; CO-05 (d)]. So "Modulo diffeomorphisms, ... the pure Ricci Flow" holds under these hypotheses. What is not mathematics: the "Mechanism" by which the agent "absorbs" the term through `Pi`, since nothing in v1.2 ties the evolution of `Pi` to φ_s (EG-02.6, item 5). v1.2's own setting ($M_{\text{model}} \subset T^*Q$, Q a Sobolev space) is outside the hypotheses (CO-02 (c)).

Repair adopted: none needed for the statement; GTPA v2 states it with the factor ½ and the hypotheses above, and does not carry the "Mechanism".

Checked: level 2 for item 1 ($2\,\mathrm{Hess}_g f - L_{\nabla f}\, g = 0$ recomputed symbolically for a general metric in two dimensions, SymPy); item 2 is a one-line computation with its reference.

En llano: lo que dice el Teorema E.1 es correcto: el término con f solo mueve las coordenadas, y deshaciendo ese movimiento lo que queda es el flujo de Ricci; que el agente lo «absorba» a través de su lente es una interpretación, no matemática.

References and sources for §3 and §4

- T. M. Cover, J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006 — Example 8.5.1.
- G. H. Golub, C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, 2013 — §8.3.
- A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002 — §2.B (invariance of domain).
- M. Li, P. Vitányi, *An Introduction to Kolmogorov Complexity and Its Applications*, 3rd ed., Springer, 2008 — §2.3.
- G. Perelman, "The entropy formula for the Ricci flow and its geometric applications", arXiv:math/0211159 (2002) — §1.
- M. Sipser, *Introduction to the Theory of Computation*, 3rd ed., Cengage Learning, 2013 — §5.1, §6.1.
- S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press, 2015 — §9.3.
- The references of the UFD nucleus (UFD §9), of Block B (§6) and of the two Block C files (§6) for the U-, B-, CT- and CO- statements cited.

Sources read: research `ff91d6f` (`ufd/UFD-NUCLEO.md`, the three block files, `gtpa-lab/CLAIM-LEDGER.md` for GTPA-C0001, C0068, C0079, C0107, C0108, C0110 and C0115, read only); house `c3406df6` (`modulos/formacion/fuentes/GTPA-v1.2.txt`, blob 32787369, for every quotation and page; FORMACION's course modules `curso/01-CALIBRACION.md` and `curso/05-CONDICIONES.md`, to see which v1.2 statements they quote).